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# Modelling large scale camera networks for identification and tracking: an abstract framework

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**Abstract:** In this study, the authors discuss a novel approach for multi-camera-based unobtrusive identification and tracking of occupants in wide-area, multi-building scenarios. Considering the scalability issues in adopting a centralised approach to monitor wide-area scenarios, they proposed a distributed approach to occupant identification and tracking. The key technical idea underlying their approach is to abstract a wide-area indoor surveillance environment using a distributed state transition system (DSTS) model, which in turn is composed of independent building-specific state transition systems, coordinating and collaborating with each other. This study presents the details of their DSTS model and examines the temporal ordering of recognition events within the DSTS for ensuring accurate state information and responses to spatio–temporal queries. They also provide an experimental evaluation of the performance of their model using precision-recall metrics. Their conclusion is that the DSTS model serves as an efficient mechanism for tracking occupants in wide-area, multi-building scenarios monitored by camera networks.

## 1 Introduction

Large-scale camera networks have been deployed in diverse application scenarios such as surveillance, smart environments, disaster management, traffic management, and event recognition [1]. Increased availability of computational resources at declining costs, rapid advancements in communication technologies and robust multimedia algorithms for analysing and drawing inferences from video and audio, has led to the commissioning of large-scale camera networks for intelligent video surveillance [1–4]. Distributed camera networks face scalability issues as the area under surveillance expands [5, 6]. The objective of this research is to provide a model for the identification and tracking of occupants within a wide-area, monitored by large-scale camera networks.

In this paper, we focus on wide-area, indoor video surveillance scenarios such as university campuses, corporate workplaces, hospitals, military bases, and elderly homes, which typically involves tracking occupants across multiple buildings. Considering the scalability issues in adopting a centralised approach to monitoring such scenarios, we extend an existing state transition system model [7] for occupant identification and tracking, by integrating the concepts from distributed systems to suit wide-area, multi-building surveillance scenarios as illustrated in Fig. 1. The movement of occupants within and across buildings in such scenarios is akin to that of mobile networks, where each cellphone is originally registered under a ‘home-network’, but could dynamically ‘roam’ in an ‘outside home-network’, where it is not originally registered; and yet continues to be reachable.

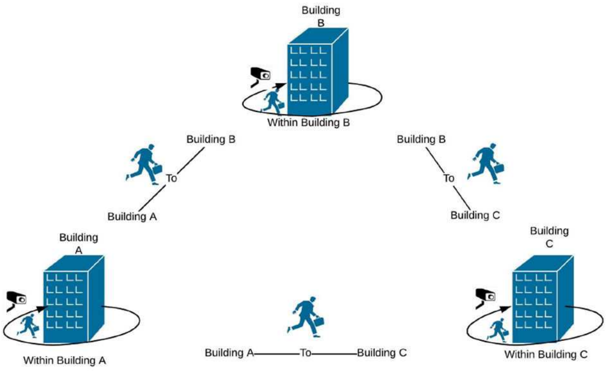


**Fig. 1** *Wide-area indoor surveillance*

The main contribution of this paper is the abstraction of a wide-area indoor surveillance environment using a distributed state transition system (DSTS) model, which in turn is composed of independent building-specific state transition systems (BSTSs), coordinating and collaborating with each other. While each BSTS is a building-specific set of states, associated events and the transition function, the DSTS encompasses a distributed set of states, distributed set of associated events and set of corresponding transition function definitions of underlying BSTS. Each state of a DSTS records the set of occupants who are present in various zones of the buildings within the wide-area environment. An event occurring in a DSTS abstracts the recognition step and the transition function abstracts the reasoning involved to effect state transitions. As the events are based on occupant probabilities generated by biometric recognition (face), the state information is expressed in terms of probabilities of the occupants being present in various zones of the buildings, as discussed in depth in [7]. The state of a DSTS changes with the occurrence of an event, i.e. the movement of an occupant within the BSTS or across the BSTS. Integrating the states of these BSTS, temporally realises a DSTS.

Another important aspect of our distributed approach for tracking is the notion of ‘occupant roaming’. Here we draw an analogy between the movement of occupants across buildings, with the domain of cellular-phone networks, where each cellphone is originally registered under a ‘home-network’, but could dynamically ‘roam’ in an ‘outside home-network’, where it is not originally registered. This roaming approach inspired by cellular-phone networks allows us to seamlessly account for occupant mobility within and across buildings. By using this approach, we effectively circumvent the complexity of state-space representation, which otherwise would have required us to maintain and continuously update a centralised mega-state table of all the buildings and associated zones. The roaming analogy helps us to realise a distributed state-space representation of a wide-area multi-building surveillance scenario.

Addressing spatio–temporal queries involving occupants who have traversed multiple zones across different buildings, such as ‘Generate the track of all occupants who visited nearby buildings’, is a non-trivial problem as the relevant state information that forms the basis for retrieval is dependent on time synchronisation and sequencing of recognition events. Considering the complexities associated with the concurrent movement of occupants across various buildings and associated state updates [8], we examine aspects of event sequencing via our DSTS model.

The rest of this paper is organised as follows. The related work is presented in Section 2, while the details of our model are discussed in Section 3, experimental evaluation in Section 4, followed by the conclusions in Section 5.

## 2 Related work

### 2.1 Camera networks for identification and tracking

Large-scale camera networks with the non-overlapping field of view serve as the main source of information for occupant identification and tracking across multiple buildings in our work. Camera networks provide potential input information for occupant identity, spatio–temporal aspects of a scene, social situation and activity recognition [9–13]. Biometric approaches to occupant identification and tracking involve modalities such as the face, gait, voice [13, 14], multimodal fusion [15], dynamic personal attributes [16]. Apart from being secure, safe and unobtrusive, face as a unique biometric modality is chosen for human identification owing to the advances in imaging devices, ease of acquiring face images and the huge availability of face images [17]. Isolating face from the captured image frame [18] and mapping it to any known face from a labelled face database are the major steps involved in face recognition [14]. Despite the advancements in face recognition technology, the recogniser accuracy deteriorates for the real-world video data due to the low resolution of images and motion blur. Additionally, external factors such as uncertainty in occupants' appearance, occlusion, changes in viewpoint, background clutter and sporadic environmental variations [14, 19] also lead to deterioration of recognition accuracy, thereby reducing the accuracy of video camera-based tracking frameworks. Besides these factors, multi-building occupant tracking can get affected by the increase in the area of surveillance and the number of occupants [4].

Multi-building occupant tracking approaches are similar to person re-identification [13] across multiple cameras present at various locations within multiple buildings of the surveillance area. An overview of the widely used techniques for human tracking using cameras is available in [4]. State-of-the-art approaches for human tracking across non-overlapping cameras can be majorly categorised as human re-identification [20, 21], camera-link model-based tracking and graph-model-based tracking [4].

Traditional surveillance techniques become inefficient with the increase in area under surveillance and duration of surveillance [2, 3]. Subsequent approaches based on intelligent video analytics addresses this situation with sophisticated computer vision algorithms. As the area under surveillance increases, performance issues related to scalability, bandwidth, security, the volume of multimedia data, and risk of a single point of failure [6], prompt an extension or replacement of centralised approaches by distributed strategies [5, 8, 22, 23]. Additionally, the presence of concurrent recognition events, inefficient multimedia data management techniques, and lack of synchronisation associated with the recognition, reasoning, and retrieval mechanisms could dampen the performance of video analytics solutions operating in wide-area [8]. Such scenarios, if left unattended can lead to a decrease in quality of service leading to erroneous identification and associated tracking results, as well as increased latency in query processing during retrieval.

Wide-area indoor surveillance applications employ a large number of visual sensors, especially cameras for monitoring occupant movement, thereby generating a massive amount of multidimensional data. A well-defined framework for distributed surveillance helps to realise an efficient design of the distributed architecture for video surveillance [1]. Hong *et al*. [8] proposed a distributed framework for large-scale situation awareness application on camera-networks, which minimises the burden on the domain experts to just providing the domain-specific handlers. Thus the ubiquitous presence of visual sensors in public and private places can form a distributed surveillance system that allows spatio–temporal situation awareness. However, such systems face various challenges caused by redundancy in multimedia data [23]. Song *et al*. [24] discussed the distributed tracking using a Kalman consensus filtering, involving hand-off between neighbouring autonomous cameras as the target moves from the field of view of one camera to the next. Scalability of such approaches to multi-building scenarios is constrained by the bandwidth requirement in inter-camera communication. A distributed inference-based scalable smart video surveillance framework as an alternative for a centralised approach by employing smart visual sensors and an overlapping field of view for occupancy reasoning is discussed in [5].

### 2.2 Review of 3R's model for identification and tracking

Our work is inspired by the 3R's model (recognition, reasoning and retrieval) [7], which proposed a unified probabilistic model using state transition systems (STS) for tracking and querying the whereabouts of occupants in smart spaces. The 3R's model elegantly abstracts the behaviour of a smart space in terms of states, events and a state transition function and transforms voluminous raw video data into a structured form for efficient information retrieval. Here a network of cameras is positioned selectively at entrances/exits/zones to realise unconstrained recognition of occupants using unobtrusive biometric modalities such as face and gait. The underlying STS model is event-based, where a 'face-recognition' event represents the transition of a registered occupant from one zone to another. Here the notion of tracking is discrete, and effectively balances privacy-versus-security concerns, and enhances occupant convenience with freedom from both tag-based and 'pause-and-declare' approaches. We summarise the salient aspects of 3R's model and focus on developing a distributed variant; and discuss the associated challenges for an expanded setting involving wide-area multi-building surveillance scenario.

The STS model can be likened to a centralised approach as the surveillance area under consideration is localised and comprises of a single building divided into multiple zones. In this approach, the state transition system $(S, E, \Delta)$ consists of states, which abstract the information necessary to support information retrieval; events, triggered by movements of occupants, abstract a biometric recognition step; and the transition function, which abstracts the reasoning behind the state transitions. This model is probabilistic in nature in order to preserve the inexactness in biometric recognition. Whenever an event occurs, the system changes its state and the corresponding state transitions from the initial state $s_0$ to final state $s_x$ following some series of events labelled as $e_1, e_2, \ldots, e_x$ may be depicted as follows:

$$s_0 \xrightarrow{e_1} s_1 \xrightarrow{e_2} s_2 \ldots \xrightarrow{e_x} s_x .$$

The transition function, which models the state change for the centralised approach is as follows:

$$\Delta: S \times E \rightarrow S \qquad (1)$$

where $S$ is the set of states, $E$ is the set of events and $\Delta$ is the transition function.

To minimise the impact of recognition errors caused by unobtrusive and unconstrained biometric modalities such as face, spatio–temporal reasoning techniques such as track-based reasoning are discussed in the 3R's model. Augmenting biometric recognition with spatio–temporal reasoning partially alleviates the shortcomings of an inefficient face recognition module and enhances the overall performance of the system. Precision and recall, standard information-theoretic metrics are modified with the use of a recognition threshold $\theta$ to evaluate the performance.

### 2.3 Shortcomings of a centralised approach

A centralised approach using the STS approach discussed in Section 2.2 handles events and state updates for a relatively smaller surveillance area, as in a single building. For instance, if we have $w$ buildings, each building with $m_u$ zones and $n_u$ registered occupants, the state information of each building can be represented in the form of a $(n_u \times m_u)$ state table, with rows representing the registered occupants and columns representing the zones. Applying the centralised approach as in [7], for modelling the multi-building surveillance scenario, generates a mega centralised state table of size $\sum_{u=1}^{w} n_u \times \sum_{u=1}^{w} m_u$ for the entire wide-area environment, which needs to be updated for every event update. A 'face-recognition' event occurring in a building due to the movement of an occupant from one zone to another, involves the generation of probabilities for all the registered occupants in the environment (including registered occupants of all other buildings) and subsequent updation of their corresponding probabilities of presence across all zones of all buildings. The centralised approach can be best understood as resulting in a mega-state table with dimensions: total number of registered occupants across all buildings (rows) × total number of zones across all buildings (columns), with each row corresponding to a registered occupant. Therefore, the centralised approach requires updation of a large monolithic state table of all the registered occupants across all buildings, for every single event, leading to scalability issues. Such a centralised approach would be rather cumbersome and unwieldy.

By employing a distributed approach, each of the building is modelled separately; with each building having a state table of size $(n_u \times m_u)$ and unlike the centralised approach, an update corresponding to the occurrence of an event is localised to the corresponding state table of the building $(n_u \times m_u)$. The distributed variant (DSTS) involving BSTS circumvents this problem by limiting each BSTS to manageable state tables of registered occupants of respective buildings. Thus for wide-area indoor surveillance scenarios with multiple buildings, a distributed approach would be more efficient than the centralised approach in terms of computational complexity of state updates.

## 3 DSTS model for wide-area indoor surveillance

Our approach employs a DSTS that involves distributed states, events, and a transition function for abstracting contextual information. The wide-area indoor surveillance environment under consideration consists of $w$-buildings, with each building $b_u$ divided into $m_u$ zones, indicating that the number of zones varies with the building. In addition to $m_u$ zones, we define the notion of building-specific transition zone denoted as $Z_{T^u}$ as the adjoining area outside a building, but within the surveillance perimeter.

### 3.1 Definitions

In this section, we provide formal definitions to the context, basic components of the DSTS, and the abstraction it provides.

*Definition (Wide-area indoor surveillance environment):* A wide-area indoor surveillance environment with $w$-buildings, forming the set $B = \{b_1, b_2, \dots, b_w\}$, is abstracted as a DSTS $(S', E', \Delta')$ where $S'$ is the set of distributed states labelled as $S^1, S^2, \dots, S^w$; $E'$ is the set of distributed events labelled as $E^1, E^2, \dots, E^w$ and $\Delta' : S' \times E' \to S'$ is the function that models a state transition on the occurrence of an event.

A building $b_u$ with $n_u$ registered occupants is abstracted as $\text{BSTS}_u$, $(S^u, E^u, \Delta^u)$ where $S^u$ is the set of building-specific states labelled as $s_0^u, s_1^u, \dots, s_{x_u}^u$; $E^u$ is the set of building-specific events labelled as $e_1^u, e_2^u, \dots, e_{x_u}^u$ and $\Delta^u : S^u \times E^u \to S^u$ is the function that models the building-specific state transition on the occurrence of a corresponding building-specific event. State transitions corresponding to building $b_u$ may be depicted as follows:

$$\text{BSTS}_u : s_0^u \xrightarrow{e_1^u} s_1^u \xrightarrow{e_2^u} s_2^u \cdots \xrightarrow{e_{x_u}^u} s_{x_u}^u$$

where as a distributed state transition that involves $w$-buildings can be depicted as follows:

$$\text{BSTS}_1 : s_0^1 \xrightarrow{e_1^1} s_1^1 \xrightarrow{e_2^1} s_2^1 \cdots \xrightarrow{e_{x_1}^1} s_{x_1}^1$$

$$\text{BSTS}_2 : s_0^2 \xrightarrow{e_1^2} s_1^2 \xrightarrow{e_2^2} s_2^2 \cdots \xrightarrow{e_{x_2}^2} s_{x_2}^2$$

$$\vdots$$

$$\text{BSTS}_w : s_0^w \xrightarrow{e_1^w} s_1^w \xrightarrow{e_2^w} s_2^w \cdots \xrightarrow{e_{x_w}^w} s_{x_w}^w .$$

The number of state transitions occurring at each BSTS varies depending upon the number of building-specific events recorded at each BSTS.

Every occupant is registered to a unique building, e.g. the one he or she might frequent on a daily basis. This is similar to the concept of 'home network' in mobile communications. Thus each building $b_u$ has a unique set of registered occupants denoted as $O_{u_R}$. It follows that the sets of registered occupants for any pair of buildings are disjoint. However, in order to model situations where registered occupants of a building visit another building (where they are not part of the list of registered occupants), we introduce the notion of a dynamic set of 'Visitors' (unregistered occupants), akin to roaming in mobile networks and denoted as $O_{u_V}$.

*Definition (Occupant):* For the $w$ -building wide-area indoor surveillance environment, the set of registered occupants $O'$, can be given as $O' = \bigcup_{u_R=1}^{w} O_{u_R}$. At any point in time, each building $b_u$ can accommodate $n_u$ occupants, from the pool of $n_{u_R}$ registered occupants (forming the set $O_{u_R} = \{o_1^{u_R}, o_2^{u_R} \dots, o_{n_{u_R}}^{u_R}\}$) and $n_{u_V}$ visitors (forming the set $O_{u_V} = \{o_1^{u_V}, o_2^{u_V} \dots, o_{n_{u_V}}^{u_V}\}$), such that for any building $b_u$ with $m_u$ zones denoted as $V_u = \{z_1^u, z_2^u, \dots, z_{m_u}^u\}$, $O_{u_R} \bigcap O_{u_V} = \varnothing$. The total occupant count within a building $b_u$ given by $n_u$ varies over time, depending on occupant movement.

For a building $b_u$ with $m_u$ zones and $n_u$ occupants, the kth state $s_k^u$ of the $\text{BSTS}_u$ is represented by $m_u$-tuple $\langle Z_{1k}^u, \dots, Z_{m_u k}^u \rangle$, were for $1 \le j \le m_u$ and $u \ne u'$

$$Z_{jk}^u = \begin{array}{l} \{\langle o_i^u, p_{jk}^u(o_i^u) \rangle \mid 1 \le i \le n_{u_R}\} \cup \\ \{\langle o_{i'}^{u'}, p_{jk}^u(o_{i'}^{u'}) \rangle \mid 1 \le i' \le n_{u_V}\} \end{array} . \tag{2}$$

In (2), $o_i^u$ corresponds to registered occupants and $o_i^{u'}$ corresponds to visitors. State $S^u$ of the $\text{BSTS}_u$ is represented as $S^u = \{s_1^u, s_2^u, \dots, s_k^u, s_{x_u}^u\}$. The state of a building is expressed in terms of the probabilities of the occupants being present in various zones of the building. The state information of a building $b_u$ is captured across two separate building-specific state tables, one for registered occupants denoted as $s\_R_{k_u}^u$ and the other for visitors of the building denoted as $s\_V_{k_u}^u$. State $s_{k_u}^u$ for a building $b_u$ can be given as follows:

$$s_{k_u}^u = \{ s\_R_{k_u}^u \cup s\_V_{k_u}^u \} . \tag{3}$$

*Definition (DSTS state):* Extending this to a wide-area indoor surveillance environment with a set of zones denoted as $V' = \bigcup_{u=1}^{w} V_u$, with a total number of zones $r = \left| \bigcup_{u=1}^{w} V_u \right|$ and $c$ occupants, where $c = \left| \bigcup_{u=1}^{w} O_{u_R} \right|$, a state $S_k'$ of the DSTS is represented as an $r$-tuple as follows:

$$\langle Z_{1k}^1, Z_{2k}^1, \dots, Z_{m_1 k}^1, Z_{1k}^2, Z_{2k}^2, \dots, Z_{1k}^w, Z_{2k}^w, \dots, Z_{m_w k}^w \rangle .$$

The state of the DSTS at the time $t$, $\langle S_k', t \rangle$ of the wide-area environment, is defined as follows:

$$\langle S'_k, t\rangle = \langle s^u_{k_u}, t'\rangle \quad \text{where}\left\{\begin{array}{c} 1 \le u \le w, t' = t, \\ (k, k_u, w)\epsilon\mathbb{N}, (t, t')\epsilon\,\mathbb{R}^+ \end{array}\right\}. \tag{4}$$

The number of events that occurred at each BSTS varies and hence $k$ and $k_u$ can take any value from $\mathbb{N}$. The overall meaningful state of the DSTS at a particular time $t$ is obtained by aggregating the local state information from each BSTS at the time $t$. The overall state of DSTS is referred to as the global state of the wide-area indoor surveillance environment. This definition of a DSTS state illustrates the importance of a common notion of time and the need for time synchronisation across the component BSTS.

However, there could be scenarios where no events were reported in the time interval $t'$ to $t$ in a BSTS, where $t' < t$. In such cases the above definition of the global state of the DSTS in (4) can be modified as follows:

$$\langle S'_k, t'\rangle = \langle s^u_{k_u}, t'\rangle \quad \text{where}\left\{\begin{array}{c} 1 \le u \le w, t' \le t, \\ \langle s^u_{k'_u}, t'\rangle \cup \langle s^u_{k_u}, t\rangle = \langle s^u_{k'_u}, t'\rangle, \\ (k, k_u, k'_u, w)\epsilon\mathbb{N}, (t, t')\epsilon\,\mathbb{R}^+ \end{array}\right\} \tag{5}$$

Among the sequence of events $\{e^u_1, e^u_2, \dots, e^u_{x_u}\}$ occurring at $\text{BSTS}_u$, the most recent event at $\text{BSTS}_u$ that occurred at a time $t'$, where $t' \le t$ will be considered towards inclusion in the DSTS state at the time $t$. If no events happened at $\text{BSTS}_u$ in time interval $[t', t]$, state information at the time $t'$ (i.e. $s^u_{k_{u'}}, t'$) will be the same as state information at the time $t$ (i.e. $s^u_{k_u}, t$).

The state of the distributed system changes with the occurrence of an event and this state information is expressed in terms of probabilities of the occupants being present in various zones of the multiple buildings. The occupant movement from one building to another happens via a transition zone, which is always outside the building, but within the area of surveillance. Each building $b_u$ has an associated transition zone labelled as $z^1_{\text{T}}, z^2_{\text{T}}, \dots, z^w_{\text{T}}$. For each state $s^u_k$, and for each occupant $o^u_i$, we enforce a constraint as follows:

$$\sum_{j=1}^{m_u} p^u_{jk}(o^u_i) + p^u_{\text{T}k}(o^u_i) = 1. \tag{6}$$

The total probability of occurrence for an occupant $o^u_i$ (registered or visitor) across all zones within a building is given by the term $\sum_{j=1}^{m_u} p^u_{jk}(o^u_i)$ and term $p^u_{\text{T}k}(o^u_i)$ gives the probability of occupant $o^u_i$ being present in the transition zone. This constraint indicates that the sum of probabilities of any occupant being present across all the zones within a building and the associated transition zone in any state equals one.

*Definition (DSTS event):* Given a $w$-building wide-area indoor surveillance environment, biometric recognition of an occupant at a particular zone is considered as an event at that zone of the building. The distance score output of the biometric recogniser can be mapped to probability distribution using the approach discussed in [7]. In the proposed work, we model recognition events as internal events that are local to each BSTS. An event $e^u_k$ local to $\text{BSTS}_u$ occurring at a time $t$ at zone $z^u_j$ $(1 \le u \le w;\ 1 \le j \le m_u)$ can be denoted as $\langle t, z^u_j, P^u_k\rangle$, where $k$ represents the order of event occurrence, $P_k = \{\langle o^u_i, p^u_{jk}(o^u_i)\rangle : 1 \le i \le n_u\}$ and $p^u_{jk}(o^u_i)$ is the probability that an occupant $o^u_i$ (registered or visitor) was recognised at the zone $j$ of building $u$ in the event $e^u_k$. $E^u$ is the set of all events associated to the $\text{BSTS}_u$, and represented as $E^u = \bigcup_{k,u\epsilon\mathbb{N}} e^u_k$. The events associated with a DSTS can be given as:

$$E' = \bigcup_{u=1}^{w} E^u. \tag{7}$$

These events are distributed across multiple BSTS. Any set of recognition events associated with a DSTS can be sequential or concurrent depending on the instance at which events occur. However, it may be noted that despite concurrency resulting from movement of occupants within and across buildings, the state transition system model effectively serialises the biometric capture event, given the availability of the exact time stamp of occurrence of the event.

*Definition (State transition function):* The function $\Delta' : S' \times E' \to S'$ maps the state $S'_{k-1}$ into state $S'_k$ upon the occurrence of an event $e^u_k$. The transition function determines the next state depending on the current state. We model each building with a BSTS and the entire area under surveillance is modelled using a DSTS. The transition function for any $\text{BSTS}_u$ can be represented as follows:

$$\Delta_u : S_u \times E_u \to S_u. \tag{8}$$

$\Delta_u$ determines local state $S_u = \langle Z^u_{1k}, \dots, Z^u_{m_u k}\rangle$ of each BSTS corresponding to each building as follows: let $x^u_i = 1 - p^u_{jk}(o^u_i)$. Then

$$Z^u_{jk} = \left\{\begin{array}{c} \langle o^u_i, p^u_{jk}(o^u_i) + x_i \times p^u_{jk-1}(o^u_i)\rangle \\ \text{where } 1 \le i \le n_u \end{array}\right\} \tag{9}$$

$$Z^u_{lk} = \left\{\begin{array}{c} \langle o^u_i, x^u_i \times p^u_{lk-1}(o^u_i) : 1 \le i \le n_u\rangle, \\ \text{for } 1 \le l \le m_u \text{ and } l \ne j \end{array}\right\}. \tag{10}$$

Information from such $w$-BSTS can be aggregated to get the information associated with the DSTS model. DSTS transition function can be depicted as follows:

$$\Delta' : S' \times E' \to S', \tag{11}$$

where $S'$ is the state of DSTS that can be deduced from (4) and (5), and $E'$ is the events associated with DSTS from (7).

### 3.2 Discussions on state table representation in DSTS

In our approach, each building is associated with a registered set of occupants who frequent the building. Movement of such occupants across other buildings in the environment, where they are not part of the registered set cannot be handled by the typical distributed state table of the building. In order to identify and track the movement of these unregistered occupants who can be considered as visitors to the building, we use an additional visitor state table. The visitor state table for the building $b_u$ has $m_u$ columns (corresponding to the number of zones in the building $b_u$) with variable number of rows depending on the number of visitors $n_{v_u}$ in that building $b_u$ at any given point in time $t$.

A state of a wide-area environment reflects the probability of presence of occupants in different zones of the various buildings in the wide-area indoor surveillance environment. In Table 1, we depict the state of a particular building $b_u$ at a particular time $t$ using a state table having $n_u$ rows (corresponding to the number of occupants in the building $b_u$) and $m_u$ columns (corresponding to the number of zones in the building $b_u$). Each BSTS has two building-specific state tables which evolve over time, based on the occurrence of events. One of these state tables is for registered occupants (Table 1) and the other for visitors (Table 2).

Table 3 illustrates an instance of the state table shown in Table 1 at the time $t_1$ for building $b_1$ with four zones, a transition zone $z^1_T$, and five registered occupants. The table shows the probability of each of the five registered occupants being present in the four zones of the building and transition zone. In this example, the registered occupant $o^1_1$ has left the building as he has been detected in the transition zone associated to building $b_1$ with a high

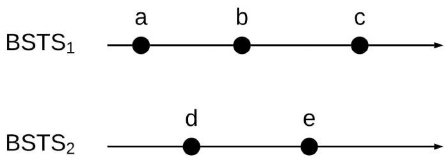


**Fig. 2** *DSTS space-time diagram*

probability, while all the other four registered occupants are within the building in zone 4.

Table 4 illustrates an instance of the state table shown in Table 2 at the time $t_2$ for building $b_2$ with three zones, a transition zone $z_T^2$, and a visitor $o_1^1$. When $o_1^1$, a registered occupant of building $b_1$ visits building $b_2$, the biometric recogniser at the entry zone $z_1^2$ of $b_2$ is unable to find a match amongst the registered occupants of $b_2$. In such scenarios, we look up a centralised master database of all the registered occupants across all the buildings and eventually identify the detected person as $o_1^1$, a registered occupant of the building $b_1$, and effectively a visitor to the building $b_2$. Though we know $o_1^1$ hails from the building $b_1$, we apply spatio–temporal reasoning to infer his arrival in $b_2$ at the time $t_2$ was from building $b_1$, given the high probability of the presence of $o_1^1$ in transition zone of building $b_1$ at the time $t_1$. Effectively, this high probability in the transition zone of the building $b_1$ confirms the absence of $o_1^1$ inside building $b_1$. As a result of this formulation, a query that checks for just the presence/absence of a registered occupant at a building level is guaranteed to retrieve an accurate result from the corresponding state table, confirming the presence/absence of the occupant. In event of the confirmed absence of a registered occupant from a building, we need to lookup the visitor tables of other adjacent buildings for presence, depending on the nature of the query in question.

### *3.3 Temporal ordering of DSTS events*

Time synchronisation and event ordering constitute some of the typical challenges associated with distributed systems [25]. In a multi-building wide-area indoor surveillance scenario, all BSTS clocks can be safely assumed to be in synchronisation and therefore the temporal order of two events could be inferred from their respective clock values. Ordering of recognition events across the BSTS and the associated state updates within each BSTS are key aspects that need to be ensured to effectively realise our DSTS model. In this discussion, we consider all the state updates to be in order if the events are in order. If events are not ordered properly, the order of state updates could get compromised and lead to erroneous tracking results. From (9) and (10), we know that current state information depends on the previous state. Since a single event occurrence updates the associated state table for the corresponding BSTS, we must ensure the ordering of events within a BSTS and at the DSTS level.

Despite the availability of timestamps associated with events, to account for concurrent movement of occupants across different buildings, we define partial ordering and total ordering of recognition events. We define 'happened before' relation for recognition events in wide-area indoor surveillance as: if $a$ and $b$ are two recognition events occurring in the same BSTS, and $a$ comes before $b$, then $a \rightarrow b$. The symbol $\rightarrow$ denotes *happen before* relation. Since $a$ happens before $b$, $a$ can causally affect $b$. If $a \rightarrow b$ and $b \rightarrow c$, then $a \rightarrow c$. Two recognition events $a$ and $b$ are said to be concurrent if, $a \nrightarrow b$ and $b \nrightarrow a$. The symbol $\|$ denotes concurrency. The concept of 'happened before' can be used to find erroneous sequence of recognition events for example an exit event recorded before an entry event.

In Fig. 2, horizontal direction represents time, and time advances forward with later time higher than earlier time. The DSTS components, BSTS are marked along the vertical direction. Each horizontal line corresponds to the advance in time for a particular BSTS. The shaded circles represent recognition events. From Fig. 2, we can infer the following: $a \rightarrow b$, $b \rightarrow c$, then $a \rightarrow c$.

**Table 1** State table for building $b_u$ at time $t$ for registered occupant $o_i^t$

| | $z_1^u$ | $z_2^u$ | $\cdots$ | $z_{m_u}^u$ | $z_T^u$ |
|---|---|---|---|---|---|
| $o_1^u$ | $p_{1k}^u(o_1^u)$ | $p_{2k}^u(o_1^u)$ | $\cdots$ | $p_{m_uk}^u(o_1^u)$ | $p_{T_uk}^u(o_1^u)$ |
| $o_2^u$ | $p_{1k}^u(o_2^u)$ | $p_{2k}^u(o_2^u)$ | $\cdots$ | $p_{m_uk}^u(o_2^u)$ | $p_{T_uk}^u(o_2^u)$ |
| $\vdots$ | $\vdots$ | $\vdots$ | $\cdots$ | $\vdots$ | $\vdots$ |
| $o_{n_u}^u$ | $p_{1k}^u(o_{n_u}^u)$ | $p_{2k}^u(o_{n_u}^u)$ | $\cdots$ | $p_{m_uk}^u(o_{n_u}^u)$ | $p_{T_uk}^u(o_{n_u}^u)$ |

**Table 2** Visitor state table entry of building $b_u$ for occupant $o_1^{u'}$ at time $t$

| | $z_1^u$ | $z_2^u$ | $\cdots$ | $z_{m_1}^u$ | $z_T^u$ |
|---|---|---|---|---|---|
| $o_1^{u'}$ | $p_{1k}^u(o_1^{u'})$ | $p_{2k}^u(o_1^{u'})$ | $\cdots$ | $p_{m_1k}^u(o_1^{u'})$ | $p_{T_1k}^u(o_1^{u'})$ |
| $\vdots$ | $\vdots$ | $\vdots$ | $\cdots$ | $\vdots$ | $\vdots$ |

**Table 3** Sample state table of registered occupants for building $b_1$

| | $z_1^1$ | $z_2^1$ | $z_3^1$ | $z_4^1$ | $z_T^1$ |
|---|---|---|---|---|---|
| $o_1^1$ | 0.020 | 0 | 0 | 0 | 0.980 |
| $o_2^1$ | 0.260 | 0.042 | 0.024 | 0.674 | 0 |
| $o_3^1$ | 0.005 | 0 | 0.032 | 0.962 | 0.001 |
| $o_4^1$ | 0.001 | 0 | 0 | 0.999 | 0 |
| $o_5^1$ | 0.128 | 0.001 | 0.042 | 0.829 | 0 |

**Table 4** Sample visitor state table depicting visitor $o_1^1$ probability for building $b_2$

| | $z_1^2$ | $z_2^2$ | $z_3^2$ | $z_T^2$ |
|---|---|---|---|---|
| $o_1^1$ | 0.999 | 0.001 | 0 | 0 |

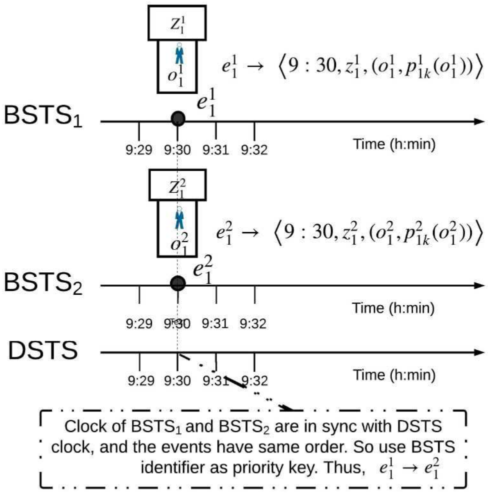


**Fig. 3** *Ordering of events in a DSTS*

Since there is no communication between the BSTS, we can assume that a pair of recognition events, happening at two different BSTS, cannot causally affect any event in other BSTSs. Thus, any event in $BSTS_1$ cannot causally affect any event in $BSTS_2$. We consider our event execution as atomic and the degree of atomicity is chosen to be the most elementary, which is occupant recognition event at each zone. Being rudimentary among the levels of atomicity ensures that a particular recognition event is indivisible and instantaneous.

As mentioned in [26], we consider clock as a means for assigning a number to a recognition event where the number is the time at which the recognition event occurred. We formally define a physical clock [26] for a wide-area indoor surveillance scenario as function $C_u(t)$, which denotes the reading of clock $C_u$ at the physical time $t$. Two events $e$ and $e'$ can be assigned clock values $k$ and $k'$, respectively if those events satisfy the below condition. For any events $e_k^u$ and $e_{k'}^{u'}$, we define the clock value assigned $k$ and $k'$ : if $e_k^u \to e_{k'}^{u'}$, then $C(e_k^u, t) < C(e_{k'}^{u'}, t') \wedge (t < t')$.

The linear ordering of events are achieved if the system satisfies the 'causal precedence' or 'happened before' property. Based on the physical clock, we define the causal precedence property [25] for recognition events happening within a BSTS as follows:

$$\forall (e_k^u, t), \forall (e_{k'}^{u'}, t') \epsilon E', e_k^u \to e_{k'}^{u'} \iff \left\{ \begin{array}{c} e_k^u \underset{u}{\to} e_{k'}^{u'}, t < t' \\ (u = u') \wedge (k < k'), \\ u, u', k, k' \epsilon \mathbb{N} \end{array} \right\}. \qquad (12)$$

Equation (12) implies that recognition event $e_k^u$ happened before the recognition event $e_{k'}^{u'}$ at the same building. Hence state update initiated by event $e_k^u$ denoted as $s_{k+1}^u$ happens before the state update initiated by event $e_{k'}^{u'}$ denoted as $s_{k'+1}^{u'}$, i.e. $s_{k+1}^u \to s_{k'+1}^{u'}$. This brings in partial ordering on the set of all events in the system.

For accurate state realisation, all events are to follow partial ordering as well as total ordering. Total ordering (relation symbol: $\Rightarrow$) for a distributed system defined in [26] is extended to a wide-area indoor surveillance as: if $e_k^u$ and $e_{k'}^{u'}$ are two recognition events happening at different zones $z_i^u$ and $z_j^u$ of $BSTS_u$, then $e_k^u \Rightarrow e_{k'}^{u'}$ if and only if either, $C_i(e_k^u) < C_i(e_{k'}^{u'})$; or $C_i(e_k^u) = C_i(e_{k'}^{u'})$ and $z_i^u \prec z_j^u$ (priority of $z_i^u$ is greater than $z_j^u$) if $i < j$.

Fig. 3 depicts ordering of two physically concurrent events $e_1^1$ occurring at $BSTS_1$ and $e_1^2$ occurring at $BSTS_2$. Since both events are having the same order of occurrence and physical time of occurrence, we can casually order the events as mentioned in Fig. 3. The above scenario of physical concurrency can be identified if and only if physical clock associated with each BSTS are in synchronisation with each other and also with DSTS and thus makes the case for physical clock and time synchronisation.

## 4 Experimental evaluation

We illustrate a realisation of the DSTS model, composed of independent BSTS by considering a college campus with a total of 35 registered occupants across two buildings. Fifteen of the occupants are registered with the building $b_1$, while the remaining 20 of them are registered with the building $b_2$. For ease of illustration, as well as to retain our focus on the distributed model, we limit the number set of zones in our experimental evaluation as $z_1^1$, $z_2^1$, $z_3^1$ and $z_T^1$ in building $b_1$ and $z_1^2$, $z_2^2$ and $z_T^2$ in the building $b_2$. We consider the movement of the ten registered occupants of the building $b_1$ over a period from 9:00 am to 4:30 pm. The remaining five registered occupants of the building $b_1$ were not present on this day. The entry of an occupant in the building $b_1$ is captured by an event and associated state transition occurring at the entry zone $z_1^1$; and the corresponding exit is captured by another event and the associated state transition in the exit zone $z_3^1$. A total 86 such events were recorded for these 10 registered occupants The transition zones $z_T^1$ and $z_T^2$ account for the probabilities of occupants while they are outside their corresponding buildings.

Face recognition has been implemented with the help of 'Face Recognition', a Python-based library built using 'Dlib' toolkit [27], which employs state-of-the-art deep learning algorithms. This face recognition module is based on a modified ResNet-34 network architecture, pre-trained on the 'Labelled Faces in the Wild' dataset with an accuracy of 99.38% [28]. Our face data set contains 40 images of each of the 35 registered occupants. The initial phase associated with the recognition task is the generation of encodings by the deep learning network for each of the faces in the registered occupants set. Each building will have its own registered set of occupants and their corresponding encodings. The entry of an occupant in a zone is captured by a camera which triggers a face recognition event with the associated state update defined by the transition function. The 'batch_face_locations' module based on CNN object detector, returns the location of all the faces within the image. These detected faces are further converted into corresponding feature vectors. The 'face_distance' module within the 'Face Recognition' library returns the distance scores upon a comparison of a detected face encoding with the encodings of registered occupants, and these distance scores get mapped to probability scores as discussed in the definition of 'Event' in Section 3. We also utilise the 'compare_faces' module available within 'Face Recognition' library, which utilises a built-in tolerance threshold to compare a list of registered occupant face encodings to a detected face encoding and determine the known face encodings that match with the detected face encoding. This will return TRUE for each match from the registered occupant set and FALSE otherwise. In our experiments, we have observed a very close correlation between the output of these two modules, i.e. customised 'face_distance' module (where customisation is limited to the additional step of mapping distance scores to probability scores) and the standard 'compare_faces' module. Of the 40 images available for each registered occupant of a building, the person with the lowest distance score (which effectively maps to the highest probability score) generated by the customised 'face_distance' module, also obtained the maximum number of 'TRUE' votes from the 'compare_faces' module. Here, the 'compare_faces' module merely serves to validate the results of the face recognition process and plays only a secondary role in our experiment. The minimum number of 'TRUE' votes that such genuine matches obtained was observed to be 12, across our experiments. In the likely event of a visitor who is not registered with a particular building, the distance scores generated will be very high as the detected face will not be similar to any of the registered occupants of the building. Additionally, the voting condition will not be satisfied in such scenarios. As a result, the recognition module will initiate a process to verify whether the individual is a visitor, i.e. a registered occupant of a nearby building, by matching against the master list of 35 registered

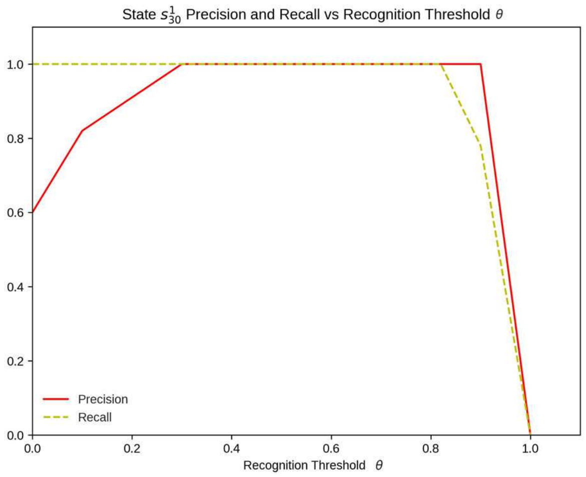


**Fig. 4** $s^1_{30}$ *Precision and recall curve*

occupants. The visitor is also confirmed based on the minimum distance score and the count for 'TRUE' votes. A recognition output, which could not be associated with a registered occupant or a visitor, is classified to be an 'Unknown' individual. Given our focus on the tracking of registered occupants across buildings, tracking of such unknown individuals is not addressed in this work, which otherwise is known to be a challenging and separate research problem by itself.

To evaluate the performance of our model, we utilise the concepts of precision and recall as defined and discussed in [7] for such contexts. The precision metric captures the extent of false positives, while the recall metric captures the extent of false negatives. These definitions are specified in terms of a recognition threshold $\theta$; implying only those people with a probability $\geq \theta$ are assumed to be present. When a person's probability in two or more zones is $\geq \theta$, the zone with the highest probability is taken as the zone of that person's presence. For a given state, the precision is the ratio of the number of occupants recognised who are in the ground truth to the total number of occupants recognised at that state. Averaging the precision across all the states leads us to the average precision. Similarly, for a given state, the recall is the ratio of the number of occupants recognised who are in the ground truth to the total number of occupants in the ground truth for that state. Averaging the recall across all states leads us to the average recall.

Table 5 illustrates the generation of state-based precision and recall metrics across varying values of the recognition threshold $\theta$. Table 5 captures a snapshot of the calculation of state-based performance metrics for varying recognition thresholds $\theta$, at the state $s^1_{30}$ of the building $b_1$, after the registered occupant $o^1_{12}$ has exited the building $b_1$ at 10:12 am by moving into the associated transition zone $z^1_{\mathrm{T}}$. Fig. 4 shows the plot of state-based precision and recall metrics for the state $s^1_{30}$. Similar calculations are repeated for each of the 86 states corresponding to the 86 events that were reported in the building $b_1$.

As can be seen in Table 5, at very low values of $\theta$, a relatively higher proportion of false positives present in the set of recognised occupants, keeps the precision low. At $\theta = 0$, $a_k = 9$, the number of occupants currently present in the building $b_1$ (as $o^1_{12}$ has already left the building and moved to transit zone $z^1_{\mathrm{T}}$), and $b_k = 15$, the total number of registered occupants in the building $b_1$, and hence precision $= a_k/b_k = 9/15 = 0.6$. The proportion of false positives present in the set of recognised occupants reduces with increasing $\theta$, until a relatively very high threshold (when $\theta$ is nearly 1), where even the true positives also fail to get recognised, leading to an overall drop in precision.

Recall, on the other hand, decreases with increasing $\theta$. As can be seen from Table 5, at low values of $\theta$, there are no false negatives present in the set of recognised occupants, thereby generating a recall value of 1. When the recognition threshold $\theta$ is increased and nears 1, even the actual occupants fail to get recognised and thus the proportion of false negatives increases, which in turn reduces the recall.

Accurate recognition results generated by the face recognition module (which are abstracted as events) has contributed to the relatively high values of probabilities of the presence for the genuine occupants observed across the sequence of states. This in turn has a direct impact on the true positives reported at high recognition thresholds, as well as limiting the false positives and false negatives. The precision and recall values for each of the 86 states are averaged to obtain average precision and average recall values for this period. These are reported in Table 6 and illustrated in Fig. 5, and are very similar to the trend for an individual state, discussed above.

The intersection of the average precision and average recall yields an optimal recognition threshold $\theta$, where the number of false positives equals the number of false negatives [7]. The overall accuracy of the face recognition module largely determines this optimal recognition threshold $\theta$. From Fig. 5, we can observe that the optimal operating point for this particular sequence of events is in the range [0.3, 0.82]. As a result, we can safely choose the higher value of 0.82 as the optimal recognition threshold for the given sequence of events. This high value for the recognition threshold is largely reflective of the overall accuracy of the underlying face recognition module, implying that a reduction in accuracy of the face recognition module will necessitate lowering

**Table 5** State-based performance metrics for state $s^1_{30}$ for varying recognition threshold $\theta$ at building $b_1$

| $\theta$ | $t_p$ | $f_p$ | $f_n$ | $a_k = t_p$ | $b_k = t_p + f_p$ | $c_k = t_p + f_n$ | Precision = $a_k/b_k$ | Recall = $a_k/c_k$ |
|---|---|---|---|---|---|---|---|---|
| 0.0 | 9 | 6 | 0 | 9 | 15 | 9 | 0.60 | 1 |
| 0.1 | 9 | 2 | 0 | 9 | 11 | 9 | 0.82 | 1 |
| 0.3 | 9 | 0 | 0 | 9 | 9 | 9 | 1 | 1 |
| 0.5 | 9 | 0 | 0 | 9 | 9 | 9 | 1 | 1 |
| 0.82 | 9 | 0 | 0 | 9 | 9 | 9 | 1 | 1 |
| 0.9 | 7 | 0 | 2 | 7 | 7 | 9 | 1 | 0.78 |
| 1 | 0 | 0 | 9 | 0 | 0 | 9 | 0 | 0 |

**Table 6** Average precision and average recall for varying $\theta$

| $\theta$ | Average precision | Average recall |
|---|---|---|
| 0.0 | 0.51 | 1 |
| 0.1 | 0.90 | 1 |
| 0.3 | 1 | 1 |
| 0.5 | 1 | 1 |
| 0.82 | 1 | 1 |
| 0.9 | 1 | 0.87 |
| 1 | 0 | 0 |

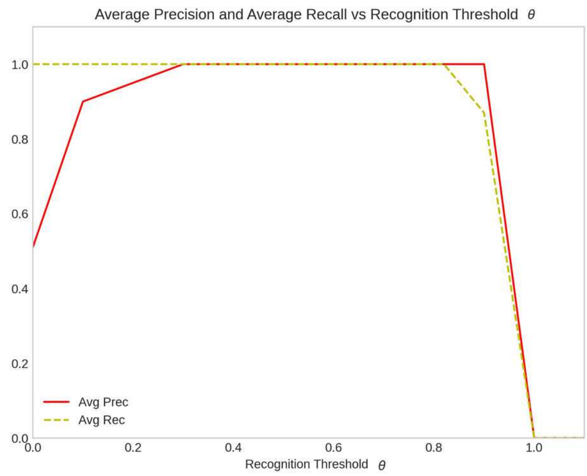


**Fig. 5** *Average precision and average recall for varying θ*

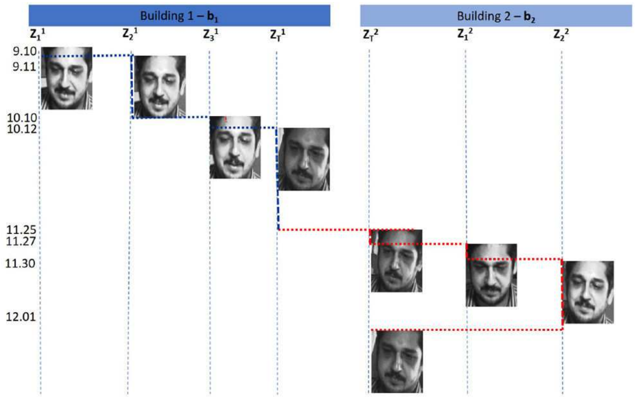


**Fig. 6** *Occupant track across buildings for occupant* $o_{12}^{1}$

of this optimal recognition threshold also. After considering multiple sequences of events involving all the registered occupants over a period of time, the optimal recognition threshold value can be fine-tuned further.

Fig. 6 illustrates the partial track of a registered occupant $o_{12}^{1}$, originally a registered occupant of the building $b_1$, who visits building $b_2$ as a visitor. The visitor track is highlighted in red.

## 5 Conclusion

In this paper, we have presented a novel approach for multi-camera-based identification and tracking in wide-area indoor surveillance scenarios. The key contributions of our paper are:

1. A DSTS model, composed of independent BSTS, coordinating and collaborating with each other, which effectively abstracts occupant identification and tracking in a multi-building wide-area indoor surveillance environment, while alleviating the shortcomings of a centralised approach.
2. An innovative approach to handle registered occupants and visitors of a given building.
3. Sequencing of recognition events in a distributed surveillance environment.

Our DSTS model integrates concepts of distributed systems, state transition systems and event-based systems for tracking occupants monitored by large scale camera networks; and is capable of addressing important aspects of a wide-area indoor surveillance system such as the sequencing of recognition events and associated state updates to ensure accurate responses to spatio–temporal queries. We have discussed an experimental evaluation of the key aspects of our DSTS model and characterised the performance of the model using precision and recall metrics. As part of our ongoing work, we intend to focus on challenges in recognition and retrieval, so as to efficiently and effectively answer spatio–temporal queries regarding the movement of occupants.

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